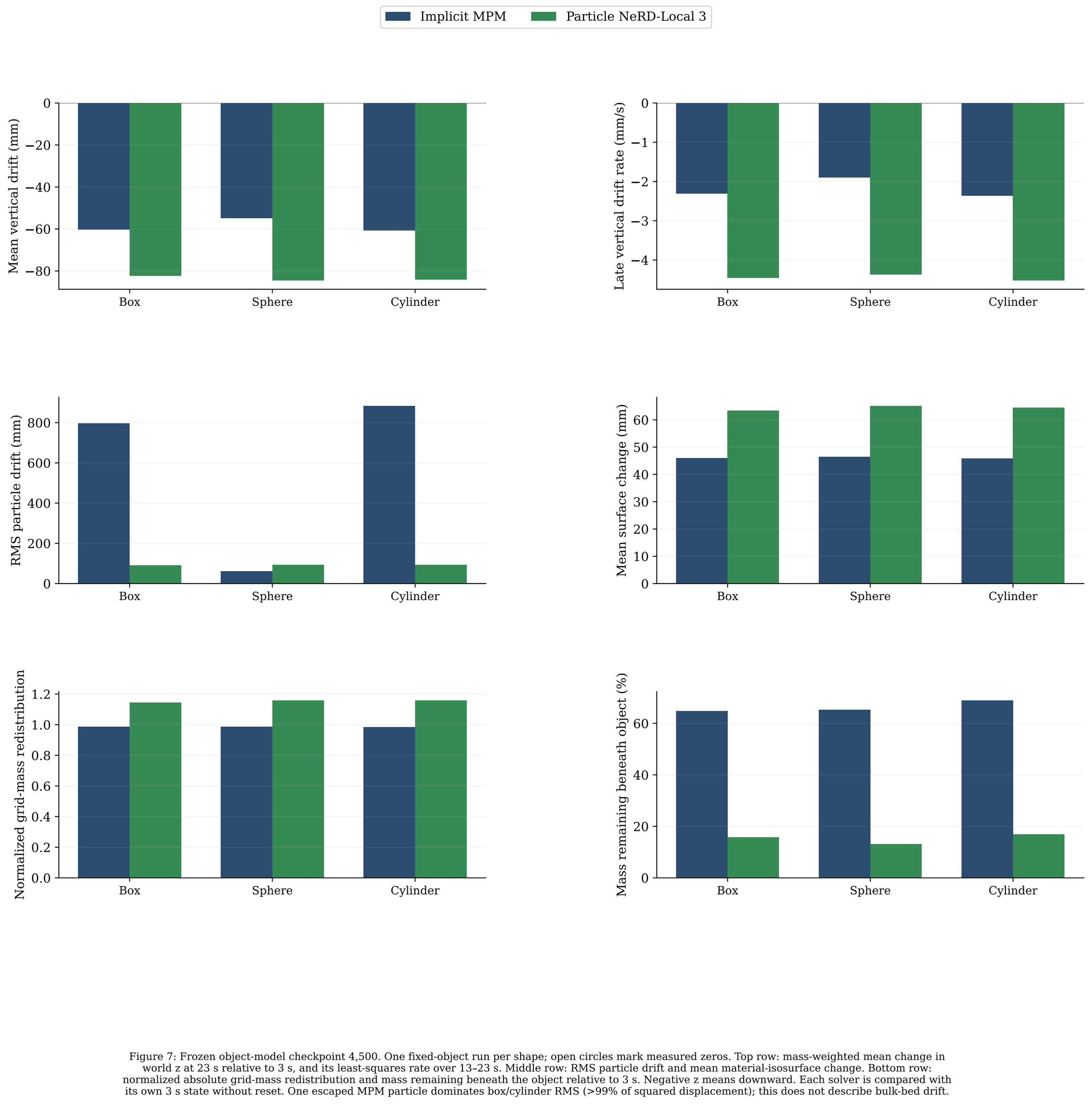

Figure 7: Frozen object-model checkpoint 4,500. One fixed-object run per shape; open circles mark measured zeros. Top row: mass-weighted mean change in world z at 23 s relative to 3 s, and its least-squares rate over 13–23 s. Middle row: RMS particle drift and mean material-isosurface change. Bottom row: normalized absolute grid-mass redistribution and mass remaining beneath the object relative to 3 s. Negative z means downward. Each solver is compared with its own 3 s state without reset. One escaped MPM particle dominates box/cylinder RMS (>99% of squared displacement); this does not describe bulk-bed drift.